Milestone 2: Summary of methodological progress and dissertation plan

Sensing Leeds Air Quality: Predictive Monitoring and Instrument Performance

MATH5872M – Dissertation in Data Science and Analytics

Student: Ali Sharifzade

Supervisors: Jim McQuaid (Faculty of Environment) and Luisa Cutillo (School of Mathematics)

1 The problem

Leeds operates a network of PurpleAir PA-II particulate sensors, spread from the city centre out to Wetherby, Otley and the rural villages east of the city. Reference-grade PM2.5 monitors are accurate but expensive, so a city normally has only a handful of them; a low-cost network trades single-instrument accuracy for spatial coverage. The question this dissertation asks is how much of the resulting data can be trusted. The sensors estimate PM2.5 optically from laser scattering rather than by weighing particles, they respond strongly to humidity, and they degrade in the field with no maintenance schedule. Some Leeds units have now been running unattended for more than four years.

The project has three aims. First, to build a cleaning and correction pipeline that turns four years of raw two-minute readings into a quality-flagged daily dataset. Second, to use the network's internal consistency, through the Concentration Similarity Index (CSI) of Byrne et al. (2024), to identify sensors behaving abnormally and produce a ranked list of candidates for physical site visits. Third, to establish whether the cleaned network is capable of resolving a genuine pollution signal at all, using a well-documented, highly localised event as an illustrative case study of the network's response rather than as a research aim in itself.

One design feature of the PurpleAir does a lot of work in what follows. Each unit contains two identical Plantower PMS5003 counters, called channels A and B, sampling the same air a few centimetres apart. Two healthy channels should agree; when they stop agreeing, at least one of them is wrong. This built-in redundancy is the foundation of the cleaning method, and its blind spot (a fault affecting both channels equally is invisible to it) is what motivates the network-level CSI.

2 Data and ingest pipeline

The raw archive is a shared OneDrive folder with one directory per sensor and one CSV per day, each holding two-minute readings of PM2.5 on both channels (CF=ATM), temperature and relative humidity. After excluding indoor units, test devices and sensors outside the study area, 48 outdoor Leeds sensors remain, contributing about 25 million rows between March 2022 and July 2026. Re-reading tens of thousands of small files on every run is impractical, so I wrote an incremental loader (src/loader.py) that keeps a manifest of file modification times and re-reads only new or changed day-files. Two details cost me real time: monthly subfolders turned out to hold exact duplicates of the daily files, and an early version of the loader silently dropped older days when switching between rebuild and append modes. I caught the latter by checking output row counts against a fresh full read, and I now treat that check as routine.

Coverage shaped the whole analysis. Most sensors report far less often than their nominal two-minute rate: across the network only about a third of hours are adequately sampled, and just one sensor would support reliable hourly averaging. Byrne et al. (2024) faced the same problem in Cork and worked at daily resolution; I do the same. A daily mean is accepted only when a sensor has at least 180 clean readings that day, about six hours of data, a threshold chosen from a sensitivity sweep as the point where little further data is lost. Sensors then need at least 100 valid days to enter the network analysis, a bar that 43 sensors clear.

3 Cleaning and humidity correction

A/B agreement flags. Following Byrne et al. (2023), every row is flagged rather than deleted, so all decisions are reversible. A reading is marked implausible if either channel exceeds the sensor's stated operating range of 1,000 μg/m³. The main flag compares the channels: their relative difference must stay within 50% at low concentrations but within 5% once the higher channel passes 15 μg/m³. The two-band structure matters because in near-clean air the counters see so few particles that large relative differences are ordinary counting noise, while at high concentrations even a small disagreement is suspicious. Just under 60% of all readings pass the flags, and Figure 1 shows why the test earns its keep: a healthy sensor's channels track each other tightly, while SL037's channel A spent months pinned at an absurd constant value that channel B never confirmed. This is precisely the hardware failure the dual-channel design exists to catch.

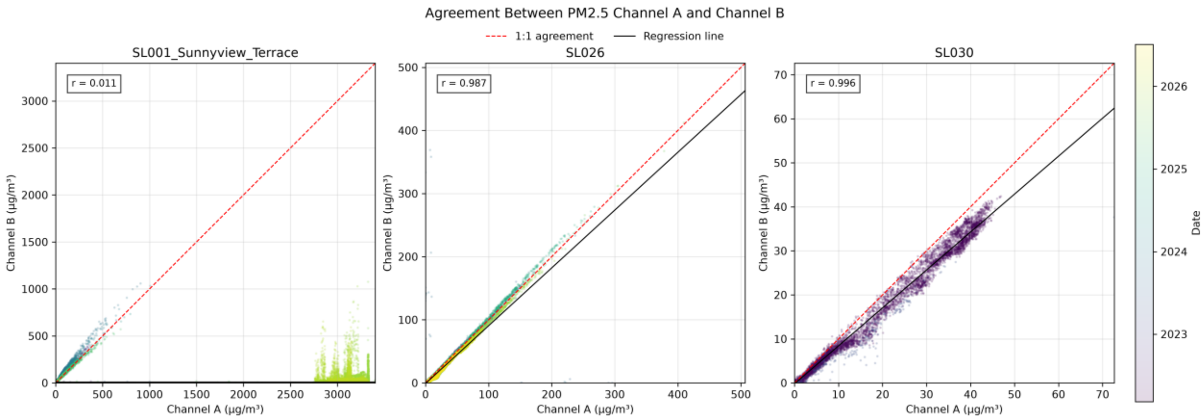

Figure 1. Channel A against channel B for three sensors. SL026 and SL030 behave as healthy instruments should, hugging the 1:1 line. SL037 (left) shows a catastrophic channel A failure: the cloud of points at the bottom right is channel A stuck near 3,300 μg/m³ while channel B reports normal air.

Humidity correction. Optical counters overestimate PM2.5 in humid air because hygroscopic particles swell. On clean rows I apply the correction of Barkjohn et al. (2021), PM = 0.524·PA − 0.0862·RH + 5.75, to the mean of the two channels, flooring at zero. Two caveats stay attached to this choice. The coefficients were fitted on United States co-location data using the sensor's CF=1 output, whereas the Leeds archive exports CF=ATM; Giordano et al. (2021) recommend CF=ATM for outdoor work, which is my justification for the pairing, but only two Leeds sensors sit anywhere near a reference-grade instrument, which is too thin a basis for a robust local verification of the correction. I therefore treat corrected values as a standardising transform that puts all sensors on a common scale rather than as an absolute calibration.

4 Network consistency: the Concentration Similarity Index

With no reference instrument available, the remaining source of truth is the network itself. The CSI of Byrne et al. (2024) compares two sensors over their common observation period: on each shared day the pair either agrees, judged by their relative difference against the geometric mean with a strict limit in polluted air and a lenient one in clean air, or it does not, and the CSI is the fraction of agreeing days. It runs from 0 (never agree) to 1 (always agree). I adapted the method in three ways, documented in src/csi.py: it is applied to daily rather than hourly means for the coverage reasons above, negative post-correction values are clipped to zero, and days where the geometric mean is zero count as agreement only when both sensors read exactly zero. Pairs need at least 30 common days to be scored, and each sensor is summarised by its mean CSI across all valid pairs.

Most of the network sits between 0.70 and 0.85, a reassuring level of coherence for instruments spread across 25 km, and pairwise CSI declines only weakly with distance, so geography alone cannot produce a very low score. The map in Figure 2 makes the fault candidates visible at a glance, and Table 1 lists the bottom of the ranking. SL051 (Bramham) is the standout case: it disagrees with almost every other sensor, and it also records the highest mean daily PM2.5 in the network despite sitting in a rural village. Elevated readings at a rural site combined with uniformly low similarity point to a positive offset fault rather than genuinely dirty air, and SL051 heads my recommended site-visit list, with its near neighbour SL050 close behind.

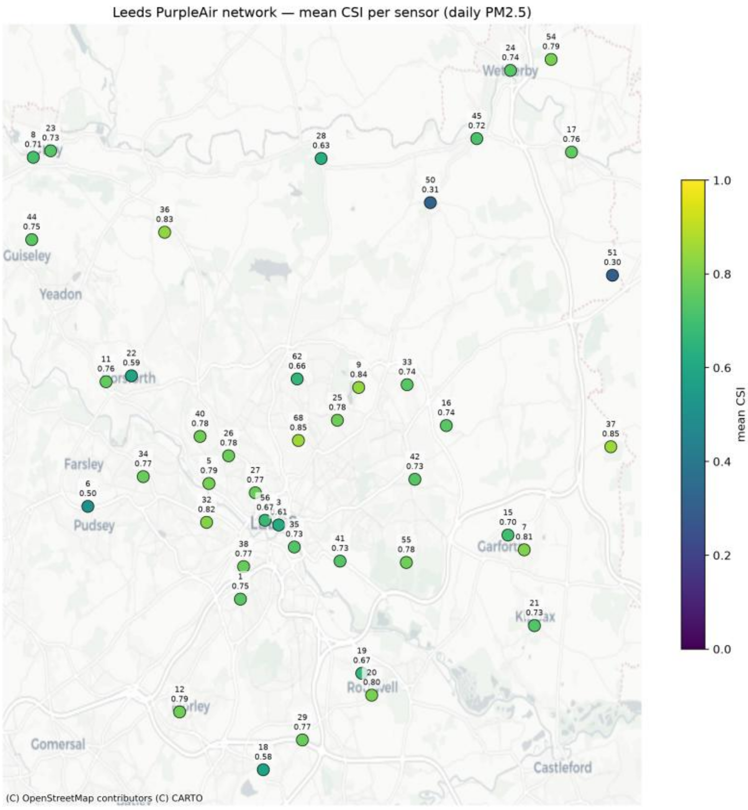

Figure 2. Mean CSI per sensor across the Leeds network on daily PM2.5. Healthy sensors (green) cluster around 0.7–0.85; the two dark blue outliers in the rural east, SL051 (Bramham) and SL050 (Bardsey), disagree with essentially the whole network and are the leading candidates for site visits.

Sensor	Mean CSI	Valid pairs
SL051 (Bramham)	0.300	37
SL050 (Bardsey)	0.312	39
SL006 (Primrose Hill Primary)	0.504	37
SL018 (East Ardsley A650)	0.581	41
SL022 (Church Lane Horsforth)	0.586	42

Table 1. The five least network-consistent sensors by mean pairwise CSI, the working site-visit list.

The most instructive result so far is the contrast between SL051 and SL037. SL037 is by far the worst sensor under the A/B test, yet on the minority of readings that survive cleaning it agrees with the network better than almost anything else. SL051 is the mirror image: its two channels agree with each other more consistently than any other sensor's, while the unit as a whole disagrees with everyone. A/B flags catch channel-level hardware divergence; the CSI catches unit-level drift or bias that both channels share. Neither test alone would have found both faults, and I intend to make this complementarity a central argument of the dissertation. I should stress that this is a first pass at network-level fault detection rather than a finished analysis: the ranking is indicative, and I expect it to sharpen once the sensitivity checks described in Section 7 have been run against the CSI thresholds and the daily-resolution adaptation.

5 Event study: Bonfire Night as a positive control

If the pipeline works, it should see fireworks: this section uses Bonfire Night purely as a check on the methodology developed in Sections 3 and 4, examining whether the cleaned data reveal a known, highly localised pollution source, rather than pursuing it as a research question in its own right. For each year from 2022 to 2025 I compared the network median PM2.5 on the evening of 5 November (17:00 to 01:00) with a control baseline, the median profile on ordinary, non-event November evenings drawn from the same years (the "control day" plotted in Figure 3), excluding 3–7 November so that neighbouring displays do not contaminate the baseline. The signal is unambiguous in all four years: evening concentrations reach seven to eleven times control levels, the composite peak lands at 20:00–21:00 exactly when displays are lit, and the plume decays through the early hours (Figure 3). Per-sensor enhancements form a plausible spatial gradient too, strongest at inner suburban sites and weakest at the rural northern edge. The network, in other words, detects a genuine multi-source urban event with the right timing, the right shape and the right geography.

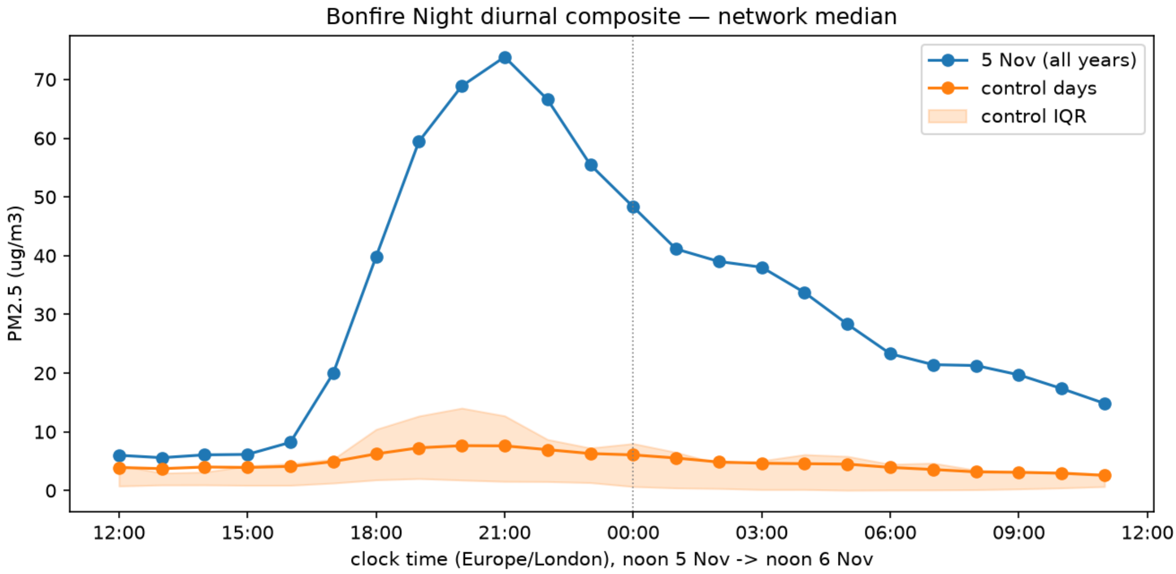

Figure 3. Bonfire Night diurnal composite of network-median PM2.5, pooled over 2022–2025, from noon on 5 November to noon on 6 November. The event evening rises an order of magnitude above the control-day baseline (orange, with interquartile shading) and peaks at 20:00–21:00.

I also examined England's 2026 World Cup matches for a hypothesised traffic-reduction effect, using per-sensor paired differences against same-weekday control days. No effect was detectable. One likely reason is the timing of the games themselves: with evening and late-evening kick-offs, most matches fell after the commuter peak, when there is little traffic left to remove. The thinner 2026 network and the small size of any plausible effect will not have helped. This null result, alongside the Bonfire Night signal, helps bracket what the network can and cannot resolve: a strong, spatially concentrated, well-timed source is clearly visible, while a diffuse and comparatively small hypothesised effect is not.

6 Tools and code organisation

All analysis is in Python (pandas, numpy, matplotlib, and contextily for basemaps), organised as a small package plus notebooks. src/config.py holds every path and threshold in one place so that sensitivity tests change a single file; src/loader.py, src/cleaning.py and src/csi.py hold the logic; notebooks 00 to 06 run the analysis from raw-data checks through cleaning and CSI to the event study. Processed data is stored as Parquet, and the code is version-controlled on GitHub with the data excluded. Each function was checked against cases with known answers before being run against the archive, and this discipline is how the loader bug and a silent resampling bug were caught.

7 Remaining work and risks

Three tasks remain before the main writing phase begins. First, I will carry out a sensitivity analysis of the CSI by testing different threshold values and evaluating the impact of the daily-resolution adaptation using modified parameters from config.py. Second, I will produce a monthly timeline of A/B sensor disagreement for each sensor to help estimate when faults first emerged, linking the analysis to the outlier-detection framework proposed by van Zoest et al. (2018). Third, I will freeze the dataset at an agreed cut-off date to ensure that the results remain consistent throughout the writing process.

In addition, I will compare the identified sensor faults with data from the nearest available reference monitoring stations, including Automatic Urban and Rural Network (AURN) sites where suitable reference data are available, to further validate the findings. I will then begin drafting the fault-detection chapter and aim to complete a draft in time for the Milestone 3 deadline in early August.

Dissertation plan

Working title: Sensing Leeds Air Quality: Predictive Monitoring and Instrument Performance

1 Introduction. Motivation for low-cost PM2.5 sensing in Leeds, the data-trust problem, and the three research aims. Closes with a summary of the dissertation's contributions.

2 Background. How optical particle counters work and the ways they fail; the PurpleAir dual-channel design; review of the quality-assurance and correction literature (Byrne et al., 2023, 2024; Barkjohn et al., 2021; Giordano et al., 2021; van Zoest et al., 2018).

3 The Leeds network and its data. Sensor inventory and the filtering to a 48-sensor study set; the incremental ingest pipeline; coverage analysis and the case for daily rather than hourly resolution.

4 Cleaning and correction. A/B agreement flagging with the two-band thresholds and their rationale; the Barkjohn humidity correction and the CF=ATM justification; per-sensor cleaning outcomes.

5 Network consistency and fault detection. The CSI, its adaptation to daily data, and the resulting sensor ranking. Argues that A/B flags and CSI catch complementary failure modes, illustrated by SL037 and SL051, and delivers a ranked site-visit list.

6 Event study. Bonfire Night 2022–2025 as a positive control, with enhancement analysis and a spatial gradient; a brief account of the England World Cup analysis, which found no detectable effect.

7 Discussion. Limitations, chiefly calibration transfer without local ground truth and the network's coverage decline through 2025–26; practical recommendations for operating the Leeds network.

8 Conclusions. Findings measured against the three aims, and directions for future work such as reference-station validation and automated fault alerts.

Appendices will contain the full code listing (with a link to the GitHub repository) and supplementary figures and tables that do not need to sit in the main body. Referencing follows the Harvard style throughout.

References

Barkjohn, K. K., Gantt, B. and Clements, A. L. (2021) 'Development and application of a United States-wide correction for PM2.5 data collected with the PurpleAir sensor', Atmospheric Measurement Techniques, 14, pp. 4617–4637.

Byrne, R., Ryan, K., Venables, D. S., Wenger, J. C. and Hellebust, S. (2023) 'Highly local sources and large spatial variations in PM2.5 across a city: evidence from a city-wide sensor network in Cork, Ireland', Environmental Science: Atmospheres, 3, doi: 10.1039/d2ea00177b.

Byrne, R., Wenger, J. C. and Hellebust, S. (2024) 'Spatial analysis of PM2.5 using a concentration similarity index applied to air quality sensor networks', Atmospheric Measurement Techniques, 17, pp. 5129–5146.

Giordano, M. R. et al. (2021) 'From low-cost sensors to high-quality data: a summary of challenges and best practices for effectively calibrating low-cost particulate matter mass sensors', Journal of Aerosol Science, 158, 105833.

van Zoest, V. M., Stein, A. and Hoek, G. (2018) 'Outlier detection in urban air quality sensor networks', Water, Air, & Soil Pollution, 229, 111.